for an exact match: if we find one, we can return current leaf as the result of search (we assume the tree doesn't contain duplicates, of course).

Listing 10.4 Method SsNode::intersectsPoint

```
function SsNode::
intersectsPoint(point)                                    #1
  return distance(this.centroid, point) <=
this.radius                          #2
```

#1 Method `intersectsPoint` is defined on `SsNode`. It takes a point and returns true if the point is within the bounding envelope of the node.

#2 Since the bounding envelope is a hyper-sphere, it just needs to check that the distance between the node's centroid and the argument point is within the node's radius. Here `distance` can be any valid metric function, including (by default) the Euclidean distance in `Rk`.

Listing 10.4 shows a simple implementation for the method checking if a point is within a node's bounding envelope; as you can see the implementation is very simple, it just uses some basic geometry. Notice, however, that the `distance` function is a structural parameter of the SS-tree; it can be the Euclidean distance in a k-dimensional space, but it can also be a different metric[8].

10.3.2 Insert